Grey Box StateSpace

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1 Outer Power Controller

1.1 Linear/Non-Linear Equations

The nonlinear equations describing the dynamics of the outer power controller are:

$$\dot{W}_P = -\left(V_{dc,ref} - V_{dc,mes}\right) \tag{1}$$

$$\dot{\Phi}_{PoC} = -\left(V_{poc,ref} - \sqrt{\left(v^c_{poc,d}\right)^2 + \left(v^c_{poc,q}\right)^2}\right) \tag{2}$$

The outputs of the block, corresponding to the current references, are defined as:

$$i_d^{ref} = k^{pu}_{i,P}\, W_P - k^{pu}_{p,P}\left(V_{dc,ref} - V_{dc,mes}\right) \tag{3}$$

$$i_q^{ref} = k^{pu}_{i,V}\, \Phi_{PoC} - k^{pu}_{p,V}\left(V_{poc,ref} - \sqrt{\left(v^c_{poc,d}\right)^2 + \left(v^c_{poc,q}\right)^2}\right) \tag{4}$$

1.2 Variables definition

Table 1: Variables of the *Outer Power Controller* model

Symbol	Description
States	
W_P	Energy state associated with active power control
Φ_{PoC}	Flux state associated with PoC voltage control
Inputs	
$V_{dc,ref}$	DC-link voltage reference
$V_{dc,mes}$	Measured DC-link voltage
$V_{poc,ref}$	PoC voltage reference
$i^c_{g,d}$	Grid current, d-axis component (controller frame)
$i^c_{g,q}$	Grid current, q-axis component (controller frame)
$v^c_{poc,d}$	PoC voltage, d-axis component (controller frame)
$v^c_{poc,q}$	PoC voltage, q-axis component (controller frame)
Outputs	
i_d^{ref}	Current reference, d-axis
i_q^{ref}	Current reference, q-axis
Parameters	
$k^{pu}_{p,P}$	Proportional gain of the active power (or DC voltage) controller
$k^{pu}_{i,P}$	Integral gain of the active power (or DC voltage) controller
$k^{pu}_{p,V}$	Proportional gain of the PoC voltage (reactive power) controller
$k^{pu}_{i,V}$	Integral gain of the PoC voltage (reactive power) controller

2 Phase-Locked Loop (PLL)

2.1 Linear/Non-Linear Equations

The nonlinear equations describing the dynamics of the PLL block are:

$$\dot{\Delta\theta} = k^{pu}_{p,PLL}\, v^{c}_{poc,q} + k^{pu}_{i,PLL}\, \Phi_{PLL} \tag{5}$$

$$\dot{\Phi}_{PLL} = v^{c}_{poc,q} \tag{6}$$

The output of the block, representing the estimated angular frequency, is given by:

$$\Delta\omega = k^{pu}_{p,PLL}\, v^{c}_{poc,q} + k^{pu}_{i,PLL}\, \Phi_{PLL} \tag{7}$$

The direct output variable of the model (used for linearization) is instead:

$$y = \Delta\theta \tag{8}$$

2.2 Variables definition

Table 2: Variables of the *Phase-Locked Loop (PLL)* model

Symbol	**Description**
States	
$\Delta\theta$	Estimated phase angle of the reference frame
Φ_{PLL}	Flux state PLL
Inputs	
$v^{c}_{poc,q}$	PoC voltage, q-axis component (controller frame)
Outputs	
$\Delta\omega$	Estimated grid angular frequency
$y = \Delta\theta$	Estimated phase angle
Parameters	
$k^{pu}_{p,PLL}$	Proportional gain of the PLL controller
$k^{pu}_{i,PLL}$	Integral gain of the PLL controller

3 Lag-Lead Compensator

3.1 Linear/Non-Linear Equations

The state equations of the *lag-lead* compensator are:

$$\dot{x} = -\frac{1}{T_2}\, x + u \tag{9}$$

$$y = \left(\frac{1}{T_2} - \frac{T_1}{T_2^2}\right) x + \frac{T_1}{T_2}\, u \tag{10}$$

corresponding to the state matrices:

$$A = -\frac{1}{T_2}, \quad B = 1, \quad C = \frac{1}{T_2} - \frac{T_1}{T_2^2}, \quad D = \frac{T_1}{T_2} \tag{11}$$

3.2 Variables definition

Table 3: Variables of the *Lag-Lead Compensator* model

Symbol	Description
Parameters	
T_1	Time constant associated with the zero pole (lead)
T_2	Time constant associated with the delayed pole (lag)
State and signal variables	
x	Internal state of the compensator
u	Input signal
y	Output signal of the compensator

4 Grid Dynamics

4.1 Linear/Non-Linear Equations

The nonlinear equations describing the grid filter dynamics are:

$$\dot{i_{g,d}}^{s} = \frac{1}{L_{grid_{pu}}}\omega_b \left(v^s_{PoC,d} - v^s_{g,d} - R_{grid_{pu}} i^s_{g,d} + \omega L_{grid_{pu}} i^s_{g,q} \right) \tag{12}$$

$$\dot{i_{g,q}}^{s} = \frac{1}{L_{grid_{pu}}}\omega_b \left(v^s_{PoC,q} - v^s_{g,q} - R_{grid_{pu}} i^s_{g,q} - \omega L_{grid_{pu}} i^s_{g,d} \right) \tag{13}$$

The outputs of the block are the stationary grid currents:

$$y = \begin{bmatrix} i^s_{g,d} \\ i^s_{g,q} \end{bmatrix} \tag{14}$$

4.2 Variables definition

Table 4: Variables of the *Grid Filter Dynamics* model

Symbol	Description
States	
$i^s_{g,d}$	Grid current d-axis (stationary frame)
$i^s_{g,q}$	Grid current q-axis (stationary frame)
Inputs	
$v^s_{PoC,d}$	Filter voltage d-axis (stationary frame)
$v^s_{PoC,q}$	Filter voltage q-axis (stationary frame)
$v^s_{g,d}$	Grid voltage d-axis (stationary frame)
$v^s_{g,q}$	Grid voltage q-axis (stationary frame)
Outputs	
$i^s_{g,d}$	Grid current d-axis (stationary frame)
$i^s_{g,q}$	Grid current q-axis (stationary frame)
Parameters	
$L_{grid_{pu}}$	Grid inductance [pu]
$R_{grid_{pu}}$	Grid resistance [pu]
ω_b	Base frequency [rad/s]
ω	Grid angular frequency [pu]

5 Frame Transformation

5.1 Linear/Non-Linear Equations

The rotation matrix is defined as:

$$T(\Delta\theta) = \begin{bmatrix} \cos\Delta\theta & \sin\Delta\theta \\ -\sin\Delta\theta & \cos\Delta\theta \end{bmatrix} \tag{15}$$

The direct transformations (stationary → controller) are:

$$\begin{aligned} \mathbf{v}_g^c &= T(\Delta\theta)\,\mathbf{v}_g^s \\ \mathbf{i}_g^c &= T(\Delta\theta)\,\mathbf{i}_g^s \\ \mathbf{i}_L^c &= T(\Delta\theta)\,\mathbf{i}_L^s \end{aligned} \tag{16}$$

where:

$$\mathbf{v}_g^s = \begin{bmatrix} v_{g,d}^s \\ v_{g,q}^s \end{bmatrix}, \quad \mathbf{v}_g^c = \begin{bmatrix} v_{g,d}^c \\ v_{g,q}^c \end{bmatrix} \tag{17}$$

The inverse transformation (controller → stationary) is:

$$\mathbf{v}_{vsc}^s = T^{-1}(\Delta\theta)\,\mathbf{v}_{vsc}^c \tag{18}$$

5.2 Variables definition

Table 5: Variables of the *Frame Transformation* model

Symbol	**Description**
	Inputs
$\mathbf{v}_g^s, \mathbf{i}_g^s, \mathbf{i}_L^s$	*dq* quantities (stationary frame)
$\mathbf{v}_{vsc}^c$	Converter voltage *dq* (controller frame)
$\Delta\theta$	Estimated phase angle (from PLL)
	Outputs
$\mathbf{v}_g^c, \mathbf{i}_g^c, \mathbf{i}_L^c$	*dq* quantities (controller frame)
$\mathbf{v}_{vsc}^s$	Converter voltage *dq* (stationary frame)

6 LCL Filter Dynamics

6.1 Linear/Non-Linear Equations

The nonlinear equations describing the LCL filter dynamics are:

$$\dot{i_{L,d}}^s = \frac{\omega_b}{L_{vsc_{pu}}} \left(v^s_{vsc,d} - v^s_{poc,d} - R_{vsc_{pu}} i^s_{L,d} + \omega L_{vsc_{pu}} i^s_{L,q}\right) \tag{19}$$

$$\dot{i_{L,q}}^s = \frac{\omega_b}{L_{vsc_{pu}}} \left(v^s_{vsc,q} - v^s_{poc,q} - R_{vsc_{pu}} i^s_{L,q} - \omega L_{vsc_{pu}} i^s_{L,d}\right) \tag{20}$$

$$\dot{v_{poc,d}}^s = \frac{\omega_b}{C_{filt_{pu}}} \left(i^s_{L,d} - i^s_{g,d} + \omega C_{filt_{pu}} v^s_{poc,q}\right) \tag{21}$$

$$\dot{v_{poc,q}}^s = \frac{\omega_b}{C_{filt_{pu}}} \left(i^s_{L,q} - i^s_{g,q} - \omega C_{filt_{pu}} v^s_{poc,d}\right) \tag{22}$$

The outputs of the block are:

$$\mathbf{y} = \begin{bmatrix} i^s_{L,d} \\ i^s_{L,q} \\ v^s_{poc,d} \\ v^s_{poc,q} \end{bmatrix} \tag{23}$$

6.2 Variables definition

Table 6: Variables of the *LCL Filter Dynamics* model

Symbol	**Description**
	States
$i^s_{L,d}, i^s_{L,q}$	LCL inductor current dq-axis (stationary frame)
$v^s_{poc,d}, v^s_{poc,q}$	PoC capacitor voltage dq-axis (stationary frame)
	Inputs
$v^s_{vsc,d}, v^s_{vsc,q}$	Converter voltage dq-axis (stationary frame)
$i^s_{g,d}, i^s_{g,q}$	Grid current dq-axis (stationary frame)
	Outputs
$i^s_{L,dq}, v^s_{poc,dq}$	Inductor current and PoC voltage dq-axis
	Parameters
$L_{vsc_{pu}}, R_{vsc_{pu}}$	Converter-side inductance and resistance [pu]
$C_{filt_{pu}}$	Filter capacitance [pu]
ω_b, ω	Base frequency [rad/s] and angular frequency [pu]

7 DC-Link Dynamics

7.1 Linear/Non-Linear Equations

The DC-link voltage dynamics is described by:

$$\dot{V}_{dc,mes} = \frac{P_n}{C_{dc} \cdot V_{dc,mes} \cdot V_{dc,n}} \left(P_{ref} - P_{meas}\right) \tag{24}$$

where the measured power is:

$$P_{meas} = v^c_{poc,d}\, i^c_{g,d} + v^c_{poc,q}\, i^c_{g,q} \tag{25}$$

The normalized output is:

$$y = \frac{V_{dc,mes}}{V_{dc,n}} \tag{26}$$

7.2 Variables definition

Table 7: Variables of the *DC-Link Dynamics* model

Symbol	Description
States	
$V_{dc,mes}$	Measured DC-link voltage
Inputs	
P_{ref}	Active power reference
$i^c_{g,d}, i^c_{g,q}$	Grid current dq-axis (controller frame)
$v^c_{poc,d}, v^c_{poc,q}$	PoC voltage dq-axis (controller frame)
Outputs	
$y = V_{dc,mes}/V_{dc,n}$	Normalized DC-link voltage [pu]
Parameters	
C_{dc}	DC capacitor capacitance [F]
$V_{dc,n}$	Nominal DC voltage [V]
P_n	Base power [W]

8 Inner Current Controller

8.1 Linear/Non-Linear Equations

The state equations of the inner current controller are:

$$\dot{C}_d = i^c_{ref,d} - i^c_{L,d} \tag{27}$$

$$\dot{C}_q = i^c_{ref,q} - i^c_{L,q} \tag{28}$$

The outputs of the controller (converter voltage commands) are:

$$v_d = \left(k^{pu}_{i,d}\, C_d + k^{pu}_{p,d}(i^c_{ref,d} - i^c_{L,d})\right) V_{dc,mes} \tag{29}$$

$$v_q = \left(k^{pu}_{i,q}\, C_q + k^{pu}_{p,q}(i^c_{ref,q} - i^c_{L,q})\right) V_{dc,mes} \tag{30}$$

8.2 Variables definition

Table 8: Variables of the *Inner Current Controller* model

Symbol	Description
States	
C_d	d-axis integrator state of the current controller
C_q	q-axis integrator state of the current controller
Inputs	
$i^c_{ref,d}, i^c_{ref,q}$	Current references dq-axis (controller frame)
$i^c_{L,d}, i^c_{L,q}$	Measured inductor currents dq-axis (controller frame)
Outputs	
v_d, v_q	Converter voltage commands dq-axis
Parameters	
$k^{pu}_{p,d}, k^{pu}_{i,d}$	Proportional/integral gains d-axis [pu]
$k^{pu}_{p,q}, k^{pu}_{i,q}$	Proportional/integral gains q-axis [pu]
$V_{dc,mes}$	Measured DC-link voltage [pu]

9 Block Connection Schema

9.1 Complete model architecture

The linear model of the VSC converter consists of 8 interconnected blocks:

Table 9: Definition of blocks and connection signals

Block	**States**	**Inputs**	**Outputs**
Power Controller	W_P, Φ_{PoC}	$V_{dc,ref}$, V_{dc,mes_filt}, $V_{poc,ref}$, $i^c_{g,dq}$, $v^c_{poc,dq}$	$i^c_{ref,d}$, $i^c_{ref,q}$
PLL	θ, Φ_{PLL}	$v^c_{poc,q}$	θ
Lag-Lead	$\Phi_{laglead}$	$V_{dc,mes}$	V_{dc,mes_filt}
Grid Filter	$i^s_{g,d}$, $i^s_{g,q}$	$v^s_{poc,dq}$, $v^s_{g,dq}$	$i^s_{g,d}$, $i^s_{g,q}$
Frame Transf.	(Static)	$v^s_{poc,dq}$, $i^s_{g,dq}$, $i^s_{L,dq}$, $v^c_{vsc,dq}$, θ	$v^c_{poc,dq}$, $i^c_{g,dq}$, $i^c_{L,dq}$, $v^s_{vsc,dq}$
LCL Filter	$i^s_{L,dq}$, $v^s_{poc,dq}$	$v^s_{vsc,dq}$, $i^s_{g,dq}$	$i^s_{L,dq}$, $v^s_{poc,dq}$
DC-Link	$V_{dc,mes}$	P_{ref}, $i^s_{g,dq}$, $v^s_{poc,dq}$	$V_{dc,mes}$
Current Ctrl.	C_d, C_q	$i^c_{ref,dq}$, $i^c_{L,dq}$	$v^c_{vsc,d}$, $v^c_{vsc,q}$